Blind Response Prediction in Reciprocal Resonator Networks
A Methods Paper and Prospective RF Test

Luqman Mahmood

30 September 2026

PREPRINT — SM4, v1.0.0
Methods and prospective protocol; no physical RF result.
DOI: 10.5281/zenodo.23049611.

Abstract

Reciprocity and an observable finite-rank modal response can determine a port response that has not been measured. We set out a prospective test using a seven-resonator RF network, two reference ports, and a previously unacquired cross-port pairing. The mathematical completion is established linear algebra; the experimental question is whether a loaded, imperfect board and its measurement chain satisfy the required rank and calibration conditions well enough to make an accurate forecast. Headline unopened transmission and acquired-but-withheld reflections are scored separately. A related, but different, synthetic reduced-array study gives sixteen successful numerical forecasts in the presence of a fixed dynamic instrument; ordinary rational interpolation agrees to numerical precision. That result supports a bounded estimator chain, not superiority over conventional circuit theory. This paper distinguishes ideal graph identities, numerical evidence, loaded-board assumptions and unperformed physical predictions.

1 What would count as a result?

The target is a complex response forecast committed before its target port pairing is acquired, with a stated uncertainty budget and refusal conditions. A physical pass would validate a bounded predictive protocol on an RF network.

The conventional comparator is the same reciprocal circuit or modal model under the same data access. A deliberately under-ranked predictor is useful as a sensitivity control, but cannot by itself establish an advantage over a competent conventional method. A fair tie remains a useful qualification result and must be reported as such. Reduced-port reconstruction of multiport scattering matrices is established metrology [1]. Low-rank skeleton completion and Nyström extension are established mathematics [3, 2]; the latter is a related real-symmetric positive-kernel setting, not an identification with complex reciprocity. Rational fitting of frequency responses, including multiport data, is an established baseline [4, 5].

2 Completion at an admitted modal rank

Partition a complex-symmetric cluster response into reference ports A and target ports B:

$$K = \begin{pmatrix} K_{AA} & K_{AB} \\ K_{BA} & K_{BB} \end{pmatrix}, \qquad K^T = K.$$

Transpose, not Hermitian conjugation, expresses this reciprocity convention.

Proposition 2.1 (Observable-rank completion)**.** *If* $\operatorname{rank} K = \operatorname{rank} K_{AA} = r$*, then*

$$K_{BB} = K_{BA} K_{AA}^{+} K_{AB}, \tag{1}$$

where $+$ *denotes the Moore–Penrose inverse. The rank condition includes the required row and column range conditions.*

Proof. Since K_{AA} has the full rank of the block matrix, the columns of K_{AB} lie in its range and the rows of K_{BA} lie in its row space. Block elimination with K_{AA}^{+} gives a Schur complement $K_{BB} - K_{BA} K_{AA}^{+} K_{AB}$. Any nonzero rank of that complement would increase the rank of K, contradicting the hypothesis. □

Noise generally destroys exact rank. The actual estimator must fix a rank-truncation rule and a conditioning cutoff before observing the target. A graph certificate alone does not prove the loaded response has that rank, or that the chosen ports see all modes.

2.1 Why unresolved clusters are the object

A nearly degenerate pair may span a well-determined subspace even when its individual residue vectors are not stably separated. We therefore target band-selected cluster residue matrices, with the band, background subtraction, normalization and pole/rank admission fixed in advance. Rank-one completion of a genuine rank-two cluster is a comparator, not an alternative definition of success. A fitted full-rank background must not be silently counted as a modal residue or discarded after the target is known.

2.2 A conditional error budget

For clarity, choose an invertible r-by-r reference block A, and write the prediction as $CA^{-1}B$. If $\sigma_{\min}(A) \geq s > 0$ and errors obey $\|\Delta A\| \leq \varepsilon_A < s$, $\|\Delta B\| \leq \varepsilon_B$ and $\|\Delta C\| \leq \varepsilon_C$, then

$$\|\widehat{C}\widehat{A}^{-1}\widehat{B} - CA^{-1}B\| \leq \frac{\varepsilon_C \|\widehat{B}\|}{s - \varepsilon_A} + \frac{\|C\|\varepsilon_A \|\widehat{B}\|}{s(s - \varepsilon_A)} + \frac{\|C\|\varepsilon_B}{s}.$$

This follows by adding three differences and using the resolvent identity. The bound is only as credible as the externally justified error inputs. It is not a perturbation theorem for a pseudoinverse across a rank change.

3 Ideal seven-resonator design versus the loaded board

The ideal comparator is the affine-E_6 seven-node star: one center, three inner nodes and three outer nodes. Its adjacency spectrum is $-2, -1, 0, 1, 2$ with multiplicities $1, 2, 1, 2, 1$. For equal inductances and compensated diagonal capacitance,

$$C = C_{\mathrm{tot}} I - C_c A, \qquad f_\lambda = \frac{f_0}{\sqrt{1 - r\lambda}}, \qquad r = C_c / C_{\mathrm{tot}}, \quad f_0 = \frac{1}{2\pi\sqrt{L C_{\mathrm{tot}}}}.$$

The design values $L = 1\,\mu\mathrm{H}$, $C_c = 68\,\mathrm{pF}$ and $C_{\mathrm{tot}} = 272\,\mathrm{pF}$ give $r = 1/4$ and ideal cluster frequencies approximately 7.879, 8.631, 9.650, 11.143 and 13.647 MHz. These are design calculations, not measured resonance frequencies.

The proposed board must be modeled with its loads. A probe capacitor in series with a termination contributes $sC_p/(1 + sZ_0C_p)$, not simply sC_p. Loss, unequal inductances, parasitics and asymmetric four-port loading alter modes, splittings and frequencies. Node-wise tuning does not make all inductances equal or restore exact ideal symmetry. The physical protocol must admit the calibrated loaded response, not require a real board to equal the ideal graph.

Two outer nodes are reference ports $P1, P2$; two inner nodes are target ports $P3, P4$. The ideal rank-two reference block has condition number three. That number is an exact ideal-design property, not a promise about the measured reference block.

4 Prospective acquisition and scoring protocol

The thresholds below are design targets; a run-specific protocol must fix them before acquisition. The supplied protocol annex lists the measurement conventions, numerical estimator and calibration decisions required before any new target capture.

1. Qualify the analyzer, complex direction convention, calibration plane, loads, component quality and board. Preserve refused admission attempts. The Touchstone test must include an asymmetric fixture: a reciprocal fixture alone cannot reveal a transposed off-diagonal writer.

2. Acquire the five reference pairings $P1P2$, $P1P3$, $P1P4$, $P2P3$ and $P2P4$. Fix the admitted bands, loaded fit, rank choices and uncertainties, and record the prediction file.

3. Commit the prediction before acquiring $P3P4$. Only S_{34} and S_{43} in that last pairing are genuinely unopened. The earlier pairings already excite $P3, P4$ and acquire their reflections.

4. Score K_{34} across the five clusters as the primary unopened prediction. Score the acquired-but-withheld reflections and the aggregate as separate consistency outcomes.

5. For center, inner and outer-node perturbations, record a new prediction before each corresponding target pairing, using the same specified regime and classification rules.

Outcome	Design relative-L^2 bound
Unopened K_{34} (headline)	≤ 0.10
Withheld self entries	≤ 0.15
Withheld aggregate	≤ 0.12

A withheld failure does not erase headline success, but must remain visible: the report must state the separate outcome of each metric. The aggregate and self metrics must use vector norms fixed in the run-specific protocol, not post-hoc elementwise tolerances.

The perturbation prediction uses the calibrated generalized capacitance pencil with a specified added 33 pF at a named node. Shifts/splittings must remain in the preselected adjacent-gap regime; outside it no orbit classification is assigned. The classification margin is a runner-up-to-best residual ratio of at least two. The attenuator-ratio check tests frequency-shape consistency only. An unchanged file or frequency-flat gain can pass it; it does not establish amplitude linearity.

4.1 Comparator addition required before a new strong claim

Alongside the rank-one control, the proposed conventional comparator is multiport vector fitting followed by reciprocal modal synthesis, using the same five reference pairings and no target-pair

data. Pole/order selection, shared-pole constraints, residue-rank rule, background, calibration, regularization and scoring must be fixed before capture. A fit that cannot identify the unseen coupling from its admitted model must report that failure, not acquire extra target information.

5 Related numerical evidence, kept in its own domain

A separate fixed-instrument synthetic intervention study produced sixteen numerical forecasts inside their declared radii [6]. The reduced arrays are supplied in `supplement/data/`, and `supplement/replay_ri7.py` reconstructs their prediction bounds. Here an *instrument* is an additive, frequency-dependent matrix response held fixed between calibration and target. A *refusal* means that the calculation withholds a forecast because its rank or fixed-instrument assumptions fail. Reported normalized prediction errors were 0.3682–1.6977 parts per million; radii were 0.01400–0.08365 percent of predicted norm. It also recorded eight rank refusals and two changing-instrument refusals. The fixed additive dynamic instrument had a matrix-norm contribution of 35.11–41.60 percent in that construction; these are not energy fractions.

Ordinary same-data rational interpolation agreed to about 5.252×10^{-16}. Thus the study corroborates a bounded numerical prediction/error chain under its specified instrument model. Its 10^{-7} complex-entry budget is a declared numerical error model.

The synthetic study's controlled-intervention rank is not automatically the modal rank in (1). The RF proposal does not reproduce that study merely by using two reference ports. The two studies motivate related questions about observability, but their assumptions and credits remain separate.

6 What a physical pass would and would not establish

A successful blinded loaded-board forecast would connect a preparation, a measured response class and a particular unseen response. Useful follow-ups include a fresh board, altered loading, controlled rank loss and a predeclared out-of-regime perturbation. Failures can locate a wrong signal class, incomplete modal observation, unstable extraction or calibration drift; these causes must be distinguished in reporting the outcome.

A future results paper must include raw complex sweeps, reference-plane metadata, prediction provenance, every refused or failed attempt and uncertainty calibration. At present there is no physical result in this manuscript.

Drafting and verification disclosure

Luqman Mahmood directed the research and is responsible for its claims and final text. OpenAI Codex and Anthropic Claude assisted mathematical work, computation, checking and editing. Statements have different evidence classes: written conditional proofs, computer-assisted certificates, numerical demonstrations, and prospective protocols. These are labelled as such in the text and do not constitute external peer review. No claim of literature priority follows from computational verification. The paper and mathematical data are licensed CC BY 4.0. Software source and executable verification scripts are excluded from that grant, with rights reserved unless a file explicitly states otherwise.

References

[1] J. C. Tippet and R. A. Speciale, *A Rigorous Technique for Measuring the Scattering Matrix of a Multiport Device with a 2-Port Network Analyzer*, IEEE Trans. Microwave Theory Tech. **30**, 661–666 (1982). https://doi.org/10.1109/TMTT.1982.1131118.

[2] C. K. I. Williams and M. Seeger, *Using the Nyström Method to Speed Up Kernel Machines*, Advances in Neural Information Processing Systems **13**, 682–688 (2001; conference 2000). https://proceedings.neurips.cc/paper/2000/hash/19de10adbaa1b2ee13f77f679fa1483a-Abstract.html.

[3] S. A. Goreinov, E. E. Tyrtyshnikov and N. L. Zamarashkin, *A theory of pseudoskeleton approximations*, Linear Algebra Appl. **261**, 1–21 (1997). https://doi.org/10.1016/S0024-3795(96)00301-1.

[4] B. Gustavsen and A. Semlyen, *Rational approximation of frequency domain responses by vector fitting*, IEEE Trans. Power Delivery **14**, 1052–1061 (1999). https://doi.org/10.1109/61.772353.

[5] Z. Drmač, S. Gugercin and C. Beattie, *Vector Fitting for Matrix-valued Rational Approximation*, SIAM J. Sci. Comput. **37**, A2346–A2379 (2015). https://doi.org/10.1137/15M1010774.

[6] L. Mahmood, Reproducibility and prospective-protocol supplement (2026). See `supplement/README.md`.